

Artist-Name Responses beyond a Shared Painting Effect in Text-to-Image Generation

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Abstract

A detectable response to a painter’s name need not reproduce differences between painters. We test this distinction in 31 digital color, spatial and texture features using 649 reproductions of Monet, Sisley, Pissarro and Cézanne. Six generators receive 14 fixed outdoor briefs under artist-free, generic-painting and four named-painter conditions, with two outputs per cell. After removing the common additive response, all six configurations have positive reference-aligned slopes, but alignment is uneven across artist pairs. GPT Image 2 nearly matches aggregate aligned amplitude ($\beta = .999$) while retaining substantial error ($D = 1.226$). FLUX.2 Max has the lowest primary error estimate and adjusted advantages over the two GPT Image 2.5 configurations; no model is established to beat the no-artist-contrast benchmark. Scalar calibration on other scenes instead gives GPT Image 2 the lowest held-scene point estimate. Source-region correction retains FLUX’s lowest point estimate but removes adjusted separation from Sunburst. The contribution is a controlled separation of name response, calibration and agreement with a specified reference collection, rather than a validated ranking of artistic fidelity.

1 Introduction

A painter’s name can change an image in several ways. It can introduce a common painting appearance, exaggerate familiar color or texture cues, or alter the features that distinguish one painter from another. These effects are easy to conflate. Moving closer to Monet’s collection, for example, does not show that a generator reproduces the differences between Monet and Sisley: a generic painting instruction could improve proximity to both.

We ask whether painter names induce differences aligned with those among *documented digital reproductions*, beyond a response shared across names. Our target is the labeled, centered mean configuration of four painters in 31 color, spatial and texture coordinates. This is an explicit collection-relative target, not a perceptually validated definition of style. Historical subject mixtures and reproduction processes can contribute to its artist differences. Holding the generated scene text fixed controls the prompt intervention; it does not make those historical subjects identical.

The distinction matters for evaluation. A generator can produce easily separable named conditions whose labels map to the wrong reference painters. It can also recover the reference-aligned amplitude while introducing large additional contrasts. Conversely, a restrained response may have lower squared error than a stronger response, without distinguishing artists more accurately in every respect. A single proximity score or recognition accuracy cannot separate these possibilities.

We cross six model configurations with identical scene descriptions and artist-free, generic oil-painting and four named-painter clauses. Centering removes their common additive shift; a projection measures reference-aligned amplitude, and an independent-repeat cross-product estimates scene-wise squared disagreement without additive sampling-noise bias. The results reveal uneven artist-pair alignment and model orderings that change with response calibration

and scene dependence. Separate post-result diagnostics examine these distinctions and reference quality. The contribution is this controlled empirical audit using established estimators.

2 Related work

Somepalli et al. (2024) develop learned style descriptors and study similarity across diffusion models. Su et al. (2025) evaluate prompted-artist recognition using content-controlled name substitutions and compare named and artist-free images with real-artist prototypes in CLIP space. Our question concerns the labeled differences among reference painters after removing a shared response; neither name substitution nor cross-model comparison is new.

Deliège et al. (2025) assess stylistic imitation and stereotyping with experts. AI-Pastiche studies authenticity judgments and prompt adherence (Asperti et al., 2025), while AI-WikiArt uses painting-derived captions and vision-language models for attribution (Fu et al., 2025). These tasks distinguish convincing imitation, recognizable authorship and distributional agreement. Here, the evaluation target is explicitly the latter in measured color, spatial and texture coordinates.

Quantitative art research studies color, composition and image statistics (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). Kim et al. (2026) combine formal and contextual representations for art-historical analysis and contextual image-to-image experiments. Our text-only name intervention supplies no original composition; its common scenes control requested generated content, not historical subjects.

Generative evaluation separates fidelity and diversity (Naeem et al., 2020), and conditional metrics separate within-condition variation from relationships among means (Benny et al., 2021). Independent-partition inner products provide noise-corrected squared differences (Diedrichsen and Kriegeskorte, 2017); energy statistics supply a complementary distributional measure (Székely and Rizzo, 2013). Processing sensitivity (Parmar et al., 2022) and limits of learned-embedding separation (Asperti, 2026) motivate the source-quality checks and the restriction to a specified digital reference target.

3 Design and measurements

3.1 Painters and reference images

The primary panel comprises 649 distinct painting reproductions: 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne. Artist means receive equal weight, so the larger Monet collection does not dominate the between-artist target. A separate development panel of 221 works supplies feature scaling; no primary reference or generated image is used to fit that transformation. Both panels were assembled before the present experiment and had been used in earlier analyses. They are explicit, finite comparison panels rather than newly withheld samples of each painter’s complete oeuvre.

Reference files carry Wikimedia Commons source metadata and Wikidata work identities (Wikimedia Commons, 2026; Vrandečić and Krötzsch, 2014). These support traceable selection and rights records, but do not standardize photography, color calibration or conservation state. The Web Gallery of Art in Kim et al. (2014), ART500K in Kim et al. (2026) and WikiArt in Fu et al. (2025) are different corpora, not validations of this collection. We therefore distinguish digital reproductions from the physical paintings.

3.2 A common-scene, six-model experiment

We compare GPT Image 1, GPT Image 2, GPT Image 2.5 Flare, GPT Image 2.5 Sunburst, Nano Banana 2 and FLUX.2 Max. Fourteen authored outdoor scenes span water, built, land and mixed compositions. Each scene is crossed with six clauses: no painting instruction; “Render as an oil painting”; and “Render as an oil painting in the style of [painter]” for each of the four painters. The scene text is otherwise identical. Each model–scene–clause cell receives two separately requested outputs, giving 1,008 images collected on 10 September 2026.

All models receive text only and return 1,024-by-1,024-pixel images. The four OpenAI models use medium quality; the other two use their specified 1K configurations. These comparisons concern the recorded configurations, not maximum attainable quality or equal expenditure per image. The exact prompts, parameters and randomized request order were fixed before collection. The 14 scenes are a fixed design, not a random sample of all possible content.

3.3 What the features measure

Images undergo orientation correction and conversion to sRGB using valid embedded profiles; compatible unprofiled files are treated as sRGB. The primary pipeline preserves aspect ratio at a 512-pixel short side, using Lanczos resizing without upsampling. No painting-region mask is applied: source margins or calibration strips can enter the measurements. Each coordinate is centered and scaled by its median and interquartile range in the development panel, with equal total weight per painter. Thus a unit is a development-panel IQR in that coordinate, not a perceptual just-noticeable difference. The Euclidean metric weights standardized coordinates equally, without further equalizing feature families.

Table 1 explains the representation. Color features summarize lightness and chroma, concentration of hue, and color changes across spatial scales; CIEDE2000 supplies the color-difference calculation (Sharma et al., 2005). Spatial features describe how luminance contrast is organized across frequencies, directions and regions. Texture features summarize multiscale detail and local intensity patterns using wavelets and local binary patterns (Mallat, 1989; Ojala et al., 2002). For example, similar mean chroma can coexist with different hue diversity, and similar total contrast can coexist with different edge orientation or fine-scale texture. These distinctions make family-specific results interpretable, while correlated coordinates and processing sensitivity prevent treating the Euclidean metric as a validated measure of artistic style.

Family	Measurements	Interpretation
Color (11)	Lightness and chroma median/IQR; chromatic fraction; hue concentration and entropy; color differences at three lags and their slope	Brightness, color strength and diversity, and how rapidly colors change over space.
Spatial (8)	Spectral slope, residual and anisotropy; orientation entropy; horizontal–vertical balance; gradient median/IQR; quadrant divergence	Coarse versus fine organization, dominant directions, edge strength and regional variation in orientation.
Texture (12)	Four wavelet energies, slope and curvature; three local-binary-pattern entropies; three local coefficients of variation	Detail across scales, diversity of local patterns and local luminance contrast.

Table 1: The 31 fixed digital features. Counts refer to scalar coordinates. Texture measurements do not measure physical paint relief or authenticate brushwork. Exact scales and definitions appear in Appendix A.

4 Measuring reference-relative artist contrasts

4.1 Reference-aligned variation

Let $\mu_a \in \mathbb{R}^{31}$ be the standardized reference mean for painter a and let $\bar{\mu} = \frac{1}{4} \sum_a \mu_a$. Define the reference painter contrasts $r_a = \mu_a - \bar{\mu}$ and their total squared magnitude $H = \sum_a \|r_a\|^2$. For generated vector z_{msak} from model m , scene s , painter a and repeat $k \in \{1, 2\}$, remove the common additive named response:

$$d_{msak} = z_{msak} - \frac{1}{4} \sum_b z_{msbk}. \quad (1)$$

Any shift shared by all four named outputs of that model, scene and repeat cancels exactly. This removes additive baselines in these coordinates. It does not remove a shared transformation: if all conditions undergo $z \mapsto Az + b$, centering removes b but leaves Ad .

The reference-aligned slope is

$$\hat{\beta}_{msk} = \frac{\sum_a \langle d_{msak}, r_a \rangle}{H}, \quad \hat{\beta}_m = \frac{1}{S} \sum_s \frac{\hat{\beta}_{ms1} + \hat{\beta}_{ms2}}{2}. \quad (2)$$

It is the least-squares coefficient of generated painter contrasts on reference contrasts. A value of zero indicates no response along the reference direction; one matches its amplitude. Values above one indicate exaggeration along that direction, not necessarily better recovery. A nonzero slope does not establish agreement for every artist pair. It is a projection onto one vectorized reference configuration, not a test that all reference contrast dimensions are recovered.

4.2 Repeat-corrected disagreement with a pooled target

Two independently requested repeats permit the cross-product estimator

$$\hat{D}_{ms} = \frac{1}{H} \sum_a \langle d_{msa1} - r_a, d_{msa2} - r_a \rangle, \quad \hat{D}_m = \frac{1}{S} \sum_s \hat{D}_{ms}. \quad (3)$$

Under stable conditional means and independent, mean-zero repeat errors, its expectation is the squared error of the scene-conditional mean painter contrasts, normalized by H . The error contributed by output sampling does not inflate this expectation. Exact agreement with these reference contrasts in every scene gives expected $D = 0$; identical conditional means for all four artist labels give expected $D = 1$. This latter benchmark is a theoretical no-contrast predictor, not an observed generic-arm estimate. Finite estimates may be negative. We do not truncate them or interpret a negative estimate as better-than-perfect recovery.

The estimator has an exact decomposition. Write $e_{msak} = d_{msak} - \hat{\beta}_{msk} r_a$. Then

$$\hat{D}_{ms} = (\hat{\beta}_{ms1} - 1)(\hat{\beta}_{ms2} - 1) + \frac{1}{H} \sum_a \langle e_{msa1}, e_{msa2} \rangle. \quad (4)$$

The first term measures reference-aligned amplitude error. The residual is orthogonal to the single stacked reference configuration; it is not necessarily outside the feature-space span of the reference means, much less outside all authentic artistic variation. Both terms can be negative in finite samples.

A simple example clarifies the target. Conditional contrasts 0, r and $2r$ give expected errors 1, 0 and 1, respectively. Permuting distinct reference labels preserves all unlabeled pairwise distances but gives nonzero error. Thus D measures labeled coordinate agreement, including response strength. It is not just a measure of separation or distance-geometry preservation.

Let θ_s stack the four conditional mean contrasts and $\bar{\theta} = S^{-1} \sum_s \theta_s$. The population target decomposes as

$$D_{\text{scene}} = \frac{\|\bar{\theta} - r\|^2}{H} + \frac{1}{SH} \sum_s \|\theta_s - \bar{\theta}\|^2 = D_{\text{aggregate}} + V_{\text{scene}}. \quad (5)$$

The second term penalizes changes in artist contrasts across scenes. Such changes need not be artistic mistakes: a generator with $\theta_s = r + u_s$ and $\sum_s u_s = 0$ matches the aggregate target but incurs positive scene-wise error. We therefore interpret the primary endpoint as *scene-wise agreement with pooled reference contrasts*. It cannot establish content-conditional artist fidelity without a corresponding historical target.

4.3 Comparing models

We fit the scene fixed-effects regression

$$\hat{D}_{ms} = \alpha_s + \gamma_m + \varepsilon_{ms}, \quad \gamma_{\text{GPT Image 2}} = 0. \quad (6)$$

All models face the same scenes and equally weighted reference painter target. With a complete balanced panel, model coefficient differences equal the mean paired scene differences. Inference uses those scene differences rather than treating features, painters or individual images as independent replicates.

The fixed inferential family contains six reference-aligned slopes and 15 pairwise model error contrasts. We report paired-scene Student intervals with $n - 1$ degrees of freedom and Bonferroni adjustment across these 21 comparisons. Nominal 95% intervals for absolute D_m are descriptive. These are approximate model-based summaries using dispersion across the fixed briefs. That dispersion mixes generation noise with genuine between-brief variation in conditional contrasts; it does not isolate fresh-request uncertainty for exactly these briefs or establish coverage for new scenes. Bonferroni adjustment requires adequately calibrated individual intervals. Two outputs permit noise correction but do not precisely characterize its sampling distribution. All 14 scenes are complete for all models.

Prespecified sensitivity analyses remove each scene in turn, resample scenes for model contrasts, and resample reference works within painter while holding generated vectors fixed. The last analysis assesses dependence on this reference panel; it is not a second independent uncertainty estimate. We also repeat the analysis in each feature family, without texture, and using central square measurement windows for both generated and reference images. The square view controls measurement-window aspect ratio while discarding peripheral content. These analyses are descriptive and do not select a preferred representation.

A separately declared sensitivity addresses unequal subject mixtures in the references. Before inspecting the new feature outcomes, we specified equal weight for four previously recorded, metadata-derived content classes within each painter, then recomputed the reference contrasts and all model estimates. For example, water scenes are much more frequent in the Monet panel than in the Pissarro panel. Equal class weighting changes the pooled mixture; it does not supply a content-conditional target, match individual compositions or identify pure style. Reference resampling within artist/class strata assesses the stability of this alternative target.

4.4 Generic controls and full distributions

The mean change from the free arm to the four named arms separates into a common component and centered painter contrasts. We compare the direction and magnitude of the common component with the free-to-generic change. Cross-repeat squared magnitudes and alignment ratios assess how much of the shared direction a generic painting clause reproduces; ordinary cosines

are retained as descriptive comparisons. A large common fraction alone cannot show that naming fails: these four reference painters also share visual properties. Equations 2–3 separately assess reference contrasts.

For painter a , energy discrepancy compares its generated collection Y_a with reference collection X_a :

$$\hat{\mathcal{E}}(X, Y) = \frac{2}{n_X n_Y} \sum_{i,j} \|x_i - y_j\| - \frac{1}{n_X^2} \sum_{i,i'} \|x_i - x_{i'}\| - \frac{1}{n_Y^2} \sum_{j,j'} \|y_j - y_{j'}\|. \quad (7)$$

Unlike distance to a mean, this statistic incorporates within-collection variation. Its finite-sample expectation depends on within-collection spread, so equal generated sample sizes do not eliminate comparison bias. We retain this declared statistic and add a correction respecting the scene-stratified design in Appendix C. Generated/reference variance ratios compare 28 outputs from 14 fixed briefs with a historically broader collection. A ratio below one is not, by itself, evidence of deficient diversity. We also report within-scene repeat spread, $\frac{1}{2S} \sum_s \|z_{msa1} - z_{msa2}\|^2$, the mean sample-variance trace of the two repeated outputs. This separates repeat dispersion from total variation across briefs.

4.5 Post-result diagnostic analyses

The declared endpoints are retained. The following analyses were added in response to review, after the primary results were known, and are descriptive. We examine artist pairs, reference singular components, label permutations, class-specific targets and disjoint real/reference controls. Subsequent checks separate control-sampling noise, correct shared-control alignment, and audit reference regions and labels. Weighting sensitivities and explicit noise simulations assess numerical stability. Appendix C specifies these retrospective analyses; full records accompany the code.

To separate contrast strength from direction, define

$$\hat{Q}_m = \frac{1}{SH} \sum_{s,a} \langle d_{msa1}, d_{msa2} \rangle, \quad \hat{D}_m = 1 - 2\hat{\beta}_m + \hat{Q}_m. \quad (8)$$

For positive $\hat{Q}_m$, $\hat{\beta}_m / \sqrt{\hat{Q}_m}$ is a noise-corrected alignment summary; finite estimates need not lie in $[-1, 1]$. Scaling every centered generated contrast by c gives $\hat{D}_m(c) = 1 - 2c\hat{\beta}_m + c^2\hat{Q}_m$. We fit $c = \max(0, \hat{\beta}/\hat{Q})$ on the other 13 scenes and evaluate it on the held-out scene. The fitting criterion is available only when its training $\hat{Q}$ is positive, as it is in every observed fold. This counterfactual concerns feature vectors, not an implemented image transformation.

5 Results

5.1 Reference-aligned responses do not establish low absolute error

All 1,008 outputs yield valid measurements. With the original full-frame reference target, every aggregate aligned slope has a simultaneous interval above zero (Table 2). This excludes a purely common additive response under the stated inference model. It does not establish agreement for every artist pair or low total error. GPT Image 2 has a slope close to one ($\beta = .999$, simultaneous interval $[.893, 1.104]$), while $D = 1.226$. Its orthogonal residual contributes 1.229; the corrected amplitude-error term is $-.003$. No amplitude-equivalence test was specified.

FLUX.2 Max has the smallest estimated primary error, .801, with a weaker aligned slope, .470. The fixed 21-comparison procedure resolves Flare minus FLUX at .937 $[.256, 1.618]$ and Sunburst minus FLUX at .840 $[.058, 1.623]$. The other 13 pairwise intervals include zero. These comparisons favor FLUX on the declared endpoint and scene panel; they do not identify a

Model	Reference slope β	Error D	ΔD versus GPT Image 2
GPT Image 1	0.938 [0.709, 1.168]	1.572	0.346 [−0.400, 1.093]
GPT Image 2	0.999 [0.893, 1.104]	1.226	0 (baseline)
GPT Image 2.5 Flare	0.772 [0.680, 0.864]	1.738	0.512 [−0.126, 1.150]
GPT Image 2.5 Sunburst	0.824 [0.680, 0.968]	1.641	0.415 [−0.350, 1.181]
Nano Banana 2	0.442 [0.256, 0.629]	1.109	−0.116 [−1.155, 0.922]
FLUX.2 Max	0.470 [0.239, 0.701]	0.801	−0.425 [−1.011, 0.161]

Table 2: Pooled-reference agreement and scene-paired model comparisons. Lower D indicates less scene-wise mismatch with pooled reference contrasts; larger β is not necessarily better. Brackets give approximate simultaneous intervals from the fixed family of six slopes and 15 model contrasts. On a complete common panel, model differences equal the scene fixed-effects regression contrasts. Inference uses paired scene differences.

uniquely best generator. Five of six error point estimates exceed the no-contrast benchmark of one, and even FLUX’s nominal interval, [.586, 1.016], includes one. No configuration is established to beat that benchmark. Absolute-error intervals were not part of the fixed inferential family.

Figure 1 illustrates the generated contrasts and their labeled reference target. These reference-only axes display 86.99% of reference-centroid variation; all numerical comparisons use 31 coordinates.

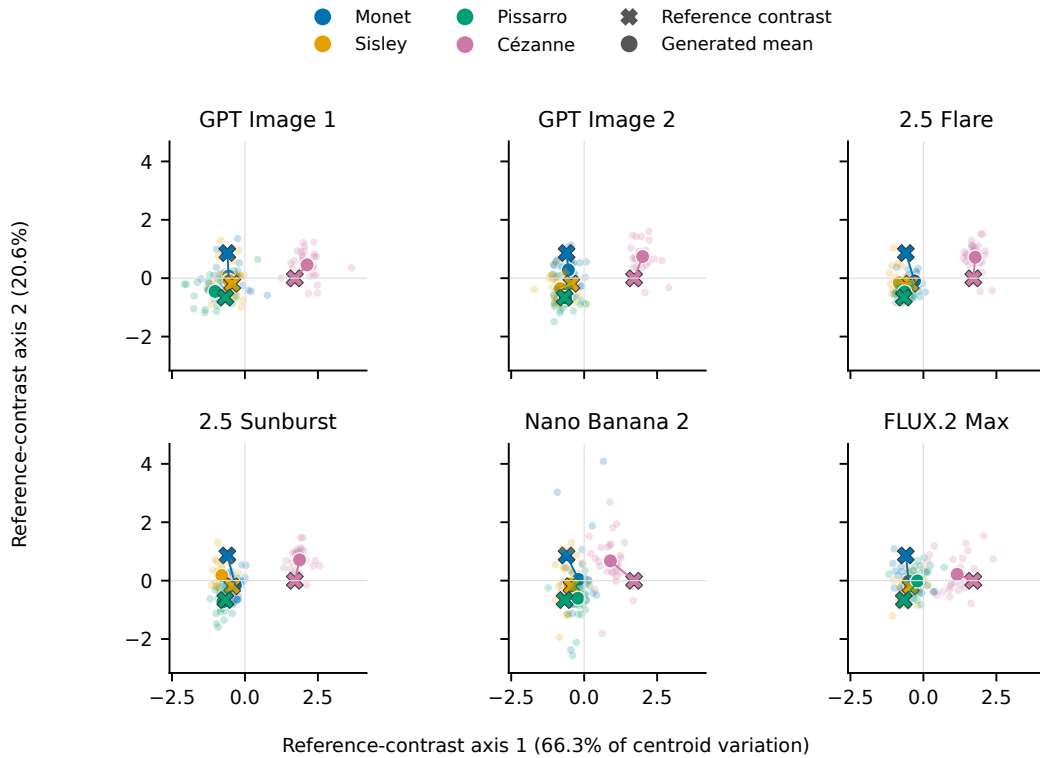


Figure 1: Centered painter contrasts. Colors identify painters; faint points are generated contrasts, circles their means, and crosses the reference contrasts. All panels share axes fitted to the four reference centroids. Separation of named conditions does not establish agreement with their labeled reference configuration or validate that configuration as a measure of artistic style.

5.2 Aggregate alignment masks uneven artist distinctions

Post-result artist-pair diagnostics show why the six positive slopes should not be interpreted as recovery of every painter distinction (Figure 2). Monet–Sisley slopes are .074 for GPT Image 1, $-.011$ for Flare, $-.249$ for Sunburst, $-.044$ for Nano Banana 2 and .129 for FLUX; GPT Image 2 has a larger estimate, .402. These are descriptive projections, not additional artist-level rejection tests.

The three nonzero reference components account for 66.3%, 20.6% and 13.0% of centroid variation. Every model has a larger slope on the dominant component than on either remaining component. Omitting Cézanne reduces FLUX’s aggregate slope from .470 to .096; the corresponding GPT Image 2 slope changes from .999 to .651. Across all 24 label assignments, the correct assignment minimizes error for GPT Image 1, GPT Image 2 and FLUX, but ranks second for Flare, Sunburst and Nano Banana 2: swapping Monet and Sisley lowers their error. This swap improvement is algebraically linked to the negative pair slope, not independent corroboration. Neither statistic establishes perceptual artist confusion.

5.3 Scene dependence and response magnitude qualify model rankings

Table 3 separates aggregate mismatch from scene variation. Nano Banana 2 has lower aggregate error than FLUX (.723 versus .761), but a larger scene-variation component (.387 versus .040), reversing the primary point ordering. This is a difference between evaluation targets; it is not evidence that all canceled scene variation is inappropriate.

Response magnitude also matters. GPT Image 2 has the highest corrected alignment ratio, .670, but generated contrast magnitude $Q = 2.223$ relative to the reference. FLUX has smaller magnitude, .741, and lower alignment, .546. A nonnegative scalar fitted on 13 scenes and evaluated on the omitted scene reduces mean error for every model. The smallest calibrated point estimate is GPT Image 2’s .554, compared with FLUX’s .714. GPT Image 2 has lower scores in 12 of 14 folds; deleting each scene and refitting the entire procedure retains its advantage,

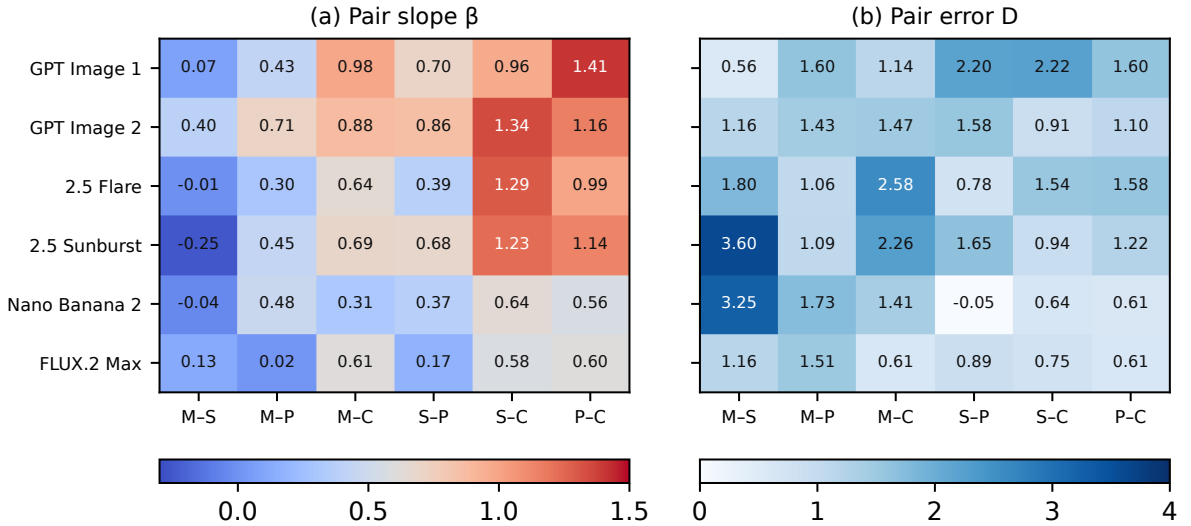


Figure 2: All six artist-pair contrasts, assessed after the primary results. M, S, P and C denote Monet, Sisley, Pissarro and Cézanne. Each pair uses its own reference difference: slope 1 matches its aligned amplitude, and expected error 1 denotes no difference between the two named conditions. Slopes and errors use both repeats and all 31 features. Positive model-level alignment can coexist with weak or reversed alignment for individual pairs.

with mean differences from $-.189$ to $-.137$. These dependent-fold sensitivities support stability within this panel, not a new superiority test or a realizable image intervention.

The observed reference title-class contrasts differ from the pooled target by an average squared amount of $.438H$, including finite-sample error and possible capture or classification differences. On the 11 unambiguously designated water, built and land briefs, FLUX retains the lowest point estimate with a class-specific target (Appendix C). These broad labels do not match compositions or verify generated scene adherence.

Genuine-painting controls separate reference disagreement from sampling noise. Across 1,000 disjoint reference half-splits, two works per painter and pseudo-scene give pooled median error $.233$ and central 95% range $[-.758, 1.380]$. Replacing those draws by each held-out pool’s exact mean narrows the range to $[.135, .428]$: much of the original width comes from sparse sampling, not changing reference geometry alone. Class-stratified controls retain larger expected errors (Appendix C). These finite-pool distributions use split-specific targets and normalizers; they are neither confidence intervals nor an absolute artistic-performance threshold.

5.4 Common shifts and distributional proximity answer other questions

The common component accounts for 82.5–95.7% of global cross-repeat-corrected named/free squared change. Computing the same decomposition within scenes before averaging gives 88.6–97.2%. Thus global pooling is not responsible for the high common fraction in this experiment. Neither fraction is a percentage of images or of artistic style. Ordinary generic/common cosines range from $.693$ to $.888$. Replacing same-repeat products with cross-repeat products avoids the shared-free-control covariance term and yields alignment ratios $.689$ – $.930$, positive after every scene deletion. The generic shift therefore shares part of the common named direction; its magnitude is smaller for every model.

Naming lowers empirical reference energy relative to the generic clause in 21 of 24 model–painter cells. A correction respecting the two-repeat scene strata retains these 21 signs. The exceptions remain Flare for Monet and Pissarro, and Sunburst for Monet. Energy and primary contrast error give different point orderings: Sunburst has the lowest named energy for Sisley and Cézanne, whereas FLUX has the lowest for Monet and Pissarro. All 24 generated/reference variance ratios are below one ($.248$ – $.877$), but these compare a fixed brief panel with broader historical collections. Complete four-painter tables and distributions appear in Appendix E.

FLUX’s point estimate remains lowest after each scene deletion and under the square-window, reference-class, family-weight and covariance sensitivities. The original square and class-weighted analyses retain both resolved comparisons; the added weighting diagnostics introduce no new tests. Removing texture leaves Flare and Sunburst errors of 1.580 and 1.604 . Persistence under these choices shows that a single texture family or broad content mixture does not explain all measured disagreement; it does not validate a unique artistic metric.

Model	$D_{\text{aggregate}}$	V_{scene}	Q	$\beta/\sqrt{Q}$	D_{held}
GPT Image 1	1.239	0.333	2.449	0.600	0.645
GPT Image 2	1.044	0.182	2.223	0.670	0.554
2.5 Flare	1.483	0.255	2.281	0.511	0.741
2.5 Sunburst	1.337	0.305	2.289	0.545	0.706
Nano Banana 2	0.723	0.387	0.994	0.444	0.832
FLUX.2 Max	0.761	0.040	0.741	0.546	0.714

Table 3: Post-result decomposition and scalar calibration. $D = D_{\text{aggregate}} + V_{\text{scene}}$, whereas $D = 1 - 2\beta + Q$ separates alignment and response magnitude. D_{held} evaluates a scalar fitted on the other 13 scenes. These are descriptive feature-space counterfactuals, without new significance tests or realizability claims.

5.5 Source correction weakens one model comparison

The source audit identifies removable peripheral regions in 90 of 649 reference and 41 of 221 development reproductions, including ten calibration-strip cases; 43 uncertain boundaries remain unchanged. Visual content assignments differ from 230 title labels, with 42 unclear cases retaining their original labels. These are assistant-coded sensitivities, not independent ground truth (Appendix C.5).

After cropping, FLUX retains the lowest error: .832 with original scaling, or .872 after refitting development scaling, compared with original .801. All six aligned slopes remain positive, and GPT Image 2 retains the lowest calibrated estimate (.579 with refitted scaling). However, the adjusted Sunburst-minus-FLUX interval now includes zero: .705 $[-.026, 1.436]$; Flare-minus-FLUX remains .864 $[.193, 1.535]$. The GPT Image 1-minus-FLUX interval also moves above zero, .758 $[.026, 1.491]$, as a retrospective sensitivity, not a new confirmatory finding. Revised visual classes on the same 11 briefs retain FLUX’s lowest point estimate. These corrections change particular comparisons while preserving the separation of alignment, magnitude and target agreement.

6 Discussion

The experiment separates reference-aligned name response, lower error on a specified target, and faithful artistic reproduction. It supports the first for all six configurations and two original comparisons under the second; the Sunburst comparison loses adjusted separation after source correction. Neither positive alignment nor a near-unit slope establishes low total error: GPT Image 2’s aligned amplitude coexists with substantial additional contrast. FLUX’s weaker response has lower uncalibrated error, while held-scene scalar calibration favors GPT Image 2 in point estimates. Both evaluations are coherent; neither defines a uniquely best artistic representation.

Aggregate alignment also conceals uneven coverage. The dominant reference component receives stronger responses than the other components, and several Monet–Sisley slopes are near zero or negative. These are weak measured pair contrasts, not evidence of perceptual confusion or an identified art-historical hierarchy. The generic clause shares part of the common named direction. Shared non-additive transformations, content substitution and reproduction processes remain possible explanations; training imbalance, preference optimization and guidance were not manipulated. The separate palette and fixed-map studies, including their contrary prospective result (Appendix F), do not establish a universal mechanism.

6.1 Scope and remaining limitations

The target is deliberately collection-relative. Scene-wise D measures consistency with pooled historical contrasts and can penalize appropriate artist-by-content variation. Coarse content classes do not match composition, viewpoint, career period or actual generated scene adherence. The source audit addresses visible peripheral regions and title/visual-class disagreements; its assistant judgments, unresolved boundaries and remaining capture differences limit the correction. A single reproduction per work cannot identify effects of illumination, sharpening, conservation or digitization source.

The coordinates measure digital properties, not physical brushwork or perceived painter resemblance. The study makes no artistic-fidelity or internal-mechanism claim. Weighting sensitivities and successful numerical replay establish neither. Likewise, the real-painting control ranges mix finite-reference disagreement with sparse sampling noise and do not define an artistic acceptance threshold.

Inference conditions on the reference panel and scaler. Simulations probe specified noise models; they cannot verify actual repeat independence. Four related painters, 14 authored

briefs, two repeats and one clause template restrict external inference. The image panels enable inspection, while the remaining gap in public exact-pixel access limits independent measurement reproduction.

7 Conclusion

Painter names produce reference-aligned digital contrasts beyond a common additive response, but the alignment is uneven across painter distinctions. Response strength, scene dependence and source handling change how configurations compare. FLUX retains the lowest error estimate after region correction, but its original adjusted Sunburst comparison does not persist; held-scene calibration gives GPT Image 2 the lowest point estimate. These findings distinguish detection, calibration and collection-relative agreement without treating any one as a validated measure of artistic fidelity.

Data and code availability

The [repository](#) contains protocols, prompts, request settings, source metadata, scalers and feature vectors. The primary analysis, diagnostic extensions and source-quality sensitivity have separate versioned records with input hashes and replay code. The source audit records all work identities, region boxes, label decisions and corrected vectors; the original results remain intact.

Appendix D displays 36 generated and four reference examples with recorded identities and byte hashes. Public numerical replay begins with vectors; independent re-extraction requires the exact source pixels, which remain outside the compact repository. Earlier experiments have [public releases](#); the current experiment and corrections still need a dedicated public archive with permissible exact pixels or a verified recovery route.

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A Feature definitions

Color coordinates use CIELAB under D65 and the 2° observer. Lightness and chroma each contribute a median and IQR. Chromatic fraction is the share of pixels with $C^* \geq 5$. Hue concentration is the magnitude of the chroma-weighted circular mean; hue entropy uses 24 equally spaced bins and is normalized by their log count. Both use chroma weights and are zero when fewer than 1% of pixels are chromatic. At each of three lags (1%, 4% and 16% of the short side), the median is taken over pooled horizontal and vertical CIEDE2000 differences. A log–log slope summarizes their scale dependence (Sharma et al., 2005).

Spatial and texture coordinates use linear-sRGB luminance. A Tukey window (parameter .1) precedes Fourier analysis. Thirty-two logarithmic radial bins cover 4–128 cycles per short side; a Theil–Sen line gives spectral slope and its root-mean-square residual. Power-weighted axial concentration measures anisotropy. Scharr derivatives supply gradient median and IQR, an 18-bin magnitude-weighted orientation entropy, and horizontal–vertical balance. Mean Jensen–Shannon divergence of quadrant orientation histograms from the global histogram measures regional variation.

Four log mean-squared detail energies come from a four-level undecimated db2 wavelet transform, followed by their linear slope and mean second difference (Mallat, 1989). Uniform local-binary-pattern entropy uses $(P, R) = (8, 1), (16, 2), (32, 4)$ on quantized luminance (Ojala et al., 2002). Three median local coefficients of variation use window widths obtained by rounding 1%, 4% and 16% of the short side and increasing even widths to the next odd integer, with a 10^{-6} mean floor. The fixed implementation records boundary handling and stabilizing constants. The separate scaler contains 101 Monet, 36 Sisley, 48 Pissarro and 36 Cézanne works. Equal painter mass is used to calculate each weighted median and IQR.

B Estimator details and sensitivity analyses

Let $\theta_{msa} = \mathbb{E}[d_{msak}]$ and $d_{msak} = \theta_{msa} + \eta_{msak}$. If repeat errors have zero mean and $\mathbb{E}\langle \eta_{msa1}, \eta_{msa2} \rangle = 0$, then

$$\mathbb{E}[\hat{D}_{ms}] = H^{-1} \sum_a \|\theta_{msa} - r_a\|^2.$$

Centering induces dependence between artists within a repeat, but independent repeat blocks preserve the vanishing cross-repeat term. A common response has $\theta_{msa} = 0$ and hence expected $D = 1$, even if it shifts the entire named collection toward the references. A correctly aligned but rescaled response $\theta_{msa} = br_a$ gives expected $D = (b - 1)^2$. Orthogonal deviations add their squared magnitude. This identity explains why stronger painter contrasts need not have lower reference-relative squared error.

For each model, global distortion applies the same cross-product estimator to the scene-averaged contrasts. Its target differs from the mean scene-wise error through the variation term in Equation 5; this term is not necessarily a failure. Both are reported, without selecting the smaller as the primary result. The primary comparison weights scenes and artists equally, irrespective of the reference count. The report also retains each painter’s additive contribution to D . These sum to the model error but are not standalone artist-fidelity scores, because the centering in Equation 1 couples the four named conditions.

For a model contrast, compute one error difference per complete paired scene. Its standard error is the sample standard deviation of those differences divided by $\sqrt{n}$. The simultaneous

half-width is $t_{n-1, 1-.05/(2.21)}$ SE. Six slope intervals use the same family adjustment. Additional nominal intervals and 5,000 paired-scene bootstrap draws are descriptive checks, not alternative routes to significance. Reference sensitivity draws 1,000 bootstrap samples independently within each painter, recomputes reference means and H , and holds generated vectors and scaling fixed. It describes reference-panel dependence rather than all uncertainty jointly.

The four-class reference sensitivity uses a fixed lexicon applied to source titles: water-organized, built-place-organized, route-organized, and open or wooded land. Each class mean receives one-quarter weight within a painter. Resampling within artist/class strata conditions on those labels and does not account for classification error. The older controlled naming panel used a separate three-class visual coding scheme, described below.

For the shared-effect diagnostic, average each arm over scenes within repeat. With named-minus-free differences h_{ak} and $c_k = \frac{1}{4} \sum_a h_{ak}$, common and specific squared magnitudes are $4\langle c_1, c_2 \rangle$ and $\sum_a \langle h_{a1} - c_1, h_{a2} - c_2 \rangle$. Their common fraction is reported when the total is positive; corrected components may be negative. Let g_k be the generic-minus-free shift. The ordinary repeat-averaged cosine shares a noisy free estimate in both vectors, adding its error covariance trace to the expected numerator. A separate diagnostic uses $[\langle g_1, c_2 \rangle + \langle g_2, c_1 \rangle] / [2\sqrt{\langle g_1, g_2 \rangle \langle c_1, c_2 \rangle}]$ when both norm squares are positive. The ratio itself is not unbiased or constrained to $[-1, 1]$. Primary D uses only named arms and is unaffected by shared-free-control noise.

C Post-result target diagnostics

These retrospective analyses retain the completed 1,008 generated images and the original inferential family. The initial diagnostics use unchanged reference vectors and scaling; the subsequent source audit separately varies painting regions, development scaling and content labels. Versioned records contain all fits, pair results, control draws, audit decisions and input hashes.

C.1 Content-specific targets and genuine-painting controls

The recorded reference classes are water-organized, built-place-organized, route-organized, and open or wooded land, assigned by a fixed source-title lexicon. Counts in that order are (191, 67, 8, 31) for Monet, (52, 29, 9, 16) for Sisley, (28, 77, 12, 24) for Pissarro and (31, 28, 12, 34) for Cézanne. Class-specific means are centered across painters just as in the pooled target. Their mean squared departure from the pooled target is $.438H$, a description of observed reference means that includes finite-sample error.

For the generated comparison, the four water, four built and three land briefs use the corresponding class means; the three mixed briefs are excluded. We divide the summed class-specific cross-product error by $H_c = S^{-1} \sum_s \sum_a \|r_{a,c(s)}\|^2 = 7.413$, compared with pooled $H = 5.915$. Each column of Table 4 therefore has its own denominator. The unchanged 11-scene pooled comparison isolates the scene-subset choice. These title-derived strata have 16–191 reference works per artist/class among the three included classes, but do not supply visually matched historical scenes.

For each of 1,000 genuine-painting controls, works in each artist/class cell are randomly divided into disjoint halves, with the smaller half used for target estimation. A pooled control samples two independent works with replacement from each painter’s held-out half for each of 14 pseudo-scenes. A class-stratified control samples within the held-out class for each of the 11 water, built or land pseudo-scenes. It is compared with both pooled and class-specific means from the training halves, using each target’s own squared magnitude. The random seed is 2026091201. Sampling can repeat a held-out work, but no held-out work is used to estimate that split’s target. The development scaler is fixed throughout.

To separate sparse sampling from changes in the comparison pools, we also calculate each split’s expected score from its exact held-out means, keeping the training target and normalizer fixed. Their central 95% ranges are $[-.135, .428]$ for pooled controls, $[-.544, .993]$ for class-stratified works against pooled targets, and $[-.517, .954]$ against class targets. The corresponding original two-draw ranges are $[-.758, 1.380]$, $[-.554, 2.400]$ and $[-.248, 1.946]$. For pooled controls, SD falls from .561 to .076 after integrating out sampling. These controls still condition on imperfect title classes and a single source capture per work; they do not calibrate artistic fidelity.

Model	11 common briefs		14 briefs	
	Pooled target	Class target	Equal family	Covariance
GPT Image 1	1.557	1.366	1.551	2.103
GPT Image 2	1.099	1.130	1.243	1.511
2.5 Flare	1.726	1.539	1.765	1.636
2.5 Sunburst	1.633	1.525	1.680	1.617
Nano Banana 2	0.987	0.948	1.203	1.385
FLUX.2 Max	0.750	0.786	0.808	0.928

Table 4: Descriptive error under alternative targets and metrics. The class target uses title-derived water, built and land strata; the three mixed briefs are excluded. Each column uses its own reference normalizer; absolute values across columns are not directly comparable. Covariance is fitted only on development works, with fixed 50% shrinkage.

Model	Reference-component slopes			Slope after omitting			
	First	Second	Third	Monet	Sisley	Pissarro	Cézanne
GPT Image 1	1.248	0.310	0.356	1.140	1.072	0.840	0.389
GPT Image 2	1.156	0.617	0.800	1.195	0.972	0.992	0.651
2.5 Flare	1.022	0.211	0.383	1.034	0.734	0.802	0.222
2.5 Sunburst	1.082	0.250	0.417	1.116	0.840	0.762	0.284
Nano Banana 2	0.513	0.371	0.195	0.569	0.446	0.388	0.276
FLUX.2 Max	0.666	0.021	0.183	0.539	0.511	0.527	0.096

Table 5: Artist coverage of the aligned response. Reference components account for 66.3%, 20.6% and 13.0% of centroid variation. Each omission recomputes the target and recenters the other three artists. These slopes are descriptive, not artist-level significance tests.

C.2 Magnitude, artist coverage and coordinate weighting

The cross-repeat version of Equation 5 is exact for the retained vectors. Its variation term replaces squared deviations with cross-products between scene-centered repeats. The term can be negative in finite samples, although its independent-repeat population target is nonnegative. Scalar calibration fits only the other 13 scenes. All fold denominators Q are positive. GPT Image 2’s fitted scalars range from .438 to .459 and FLUX’s from .598 to .694; all fold-specific scores are retained. No calibration parameter is selected by minimizing its own held-out score. The subsequent deletion check refits the entire procedure on each remaining 13-scene panel, rather than treating overlapping folds as independent replications.

For artist pairs, both generated and reference contrasts are direct differences between the two artists; the denominator is the squared reference-pair distance. Leaving one artist out recenters the remaining three generated and reference conditions. Reference singular components are the three rank-one terms of the centered four-by-31 reference matrix: each combines an artist contrast with a feature direction, rather than representing one artist. Slopes project onto each term separately. Table 5 shows that aggregate positive alignment is uneven across those components. The 24 permutations rearrange generated artist labels while leaving the

reference fixed. Their ranks are not exact permutation p -values: the named conditions are not exchangeable under an established null.

Equal-family weighting gives each coordinate a weight inversely proportional to its family’s size (11, 8 or 12). The covariance sensitivity uses the 221 development works with equal total mass per painter. For their weighted covariance Σ , the metric matrix is $[\.5\Sigma + \.5\text{tr}(\Sigma)I/31]^{-1}$; the shrinkage is fixed, without tuning on the evaluated images. These changes address correlated coordinate weighting but are not independently validated artistic metrics. FLUX retains the smallest point estimate under each of 31 single-coordinate deletions. Every coordinate contribution, including negative terms, is retained in the report rather than selecting a favorable subset.

C.3 A design-aware energy correction

The primary energy values use empirical distributions, including their zero diagonal distances. To estimate energy between a fixed empirical reference and the equally weighted mixture of scene-conditional output distributions, cross-scene generated pairs already have the required independent draws. Only the same-scene generated term needs adjustment. With two repeats and S scenes, the descriptive correction is

$$\hat{\mathcal{E}}_{\text{strata}} = \hat{\mathcal{E}}_{\text{empirical}} - \frac{1}{2S^2} \sum_s \|y_{s1} - y_{s2}\|.$$

The reference within-distance remains empirical because this target conditions on the finite reference panel. A pooled IID $n/(n-1)$ multiplier would also alter cross-scene terms and estimate a different target. This correction retains the 21 negative named-minus-generic differences. It does not solve unequal historical and generated content mixtures, and no new energy tests are performed.

C.4 Uncertainty and synthetic coverage checks

The Student procedure uses 14 scene summaries. For independent scene errors with fixed conditional means δ_s , its estimated variance has expectation

$$\mathbb{E}[s_\delta^2/S] = \text{Var}(\bar{\delta}) + \frac{1}{S(S-1)} \sum_s (\delta_s - \bar{\delta})^2.$$

It therefore includes real between-scene heterogeneity as well as request noise. This explains why it is not a direct estimate of fresh-request variance for exactly the fixed panel. It also does not provide an exact coverage guarantee: non-Gaussian cross-products and dependence can affect the Student approximation.

The initial 5,000-run scenarios (seed 2026091202) use the six configurations, 14 scenes, two repeats and 21-endpoint family, with unit reference magnitude. Gaussian, variance-standardized t_3 /heteroskedastic and fixed scene-interaction errors give simultaneous coverage 95.96%, 97.42% and 99.68%. Their independent centered noise power is .5. Allocating half the noise to a model-specific state shared by all scenes and both repeats reduces coverage to 0.04%, with mean corrected-error bias near .25.

A subsequent sweep uses independent Gaussian errors at powers .5, 1.5 and 3, with either isotropic or rank-three covariance aligned with the reference components (5,000 runs per case). Coverage ranges from 95.06% to 96.80%, with Monte Carlo SE .25–.31 percentage points. For comparison, directly centered repeat differences estimate observed powers of .424–2.596 relative to H . Full noise definitions and results are retained in the versioned report. These stress models probe coverage approximations, not actual service independence; no significance threshold is selected from them.

C.5 Source-quality sensitivity

An assistant visual audit inspects all 649 reference and 221 development reproductions using numbered contact sheets and enlarged peripheral regions. It records visible calibration targets, photographic frames, margins and labels, with normalized rectangular painting-region coordinates. Ambiguous boundaries remain unchanged. Reference content is assigned to the four broad classes from the image before comparison with the original title label; mixed or unclear cases retain that original label. This is source inspection, not a human artist-resemblance evaluation or composition matching.

For clearly bounded regions, the same orientation and ICC rules precede crop and aspect-preserving Lanczos resizing to a 512-pixel short side without upsampling. All 31 feature definitions and all works are retained. We compare corrected references first with the original scaler, then with IQR scaling refitted on corrected development images and consistently applied to generated vectors. A separate comparison changes content labels on the same 11 briefs. The audit and new compact vectors are versioned alongside the original results; corrections are not chosen by optimizing model scores.

D Inspectable image examples

The generated panel uses the first declared brief and first repeat for every model and clause, selected by this fixed rule after numerical analysis. The brief reads: “A broad river bends past a gravel bank. Three tall trees stand on the far bank, with a low hill behind them under a pale overcast sky.” Every prompt ends with “No text or frame.” Artist-free, generic and named clauses otherwise follow Section 3. Images were not selected by visual success, distance or agreement with the paper’s claims.

Reference examples are the lexicographically first measured water-class work identifiers with recorded public-domain status under the original title labels. The Sisley and Pissarro reproductions retain calibration strips; the Cézanne example, *Crossroad of Rue Rémy, Auvers*, depicts a street. These original full-frame examples document the defects motivating the separate source audit (Appendix C.5); they are not matched depictions of the generated brief. Source identifiers, exact requests and byte hashes accompany the panels, whose reduced images support inspection rather than re-extraction.



Figure 3: Rule-selected reference examples. Columns identify painters and recorded Wikidata work identifiers. Calibration strips are visible beside the Sisley and Pissarro reproductions; the Cézanne example reveals a title-class mismatch. Source URLs and public-domain metadata are retained with the inspection manifest. The works are not exact content matches to the common generated brief.

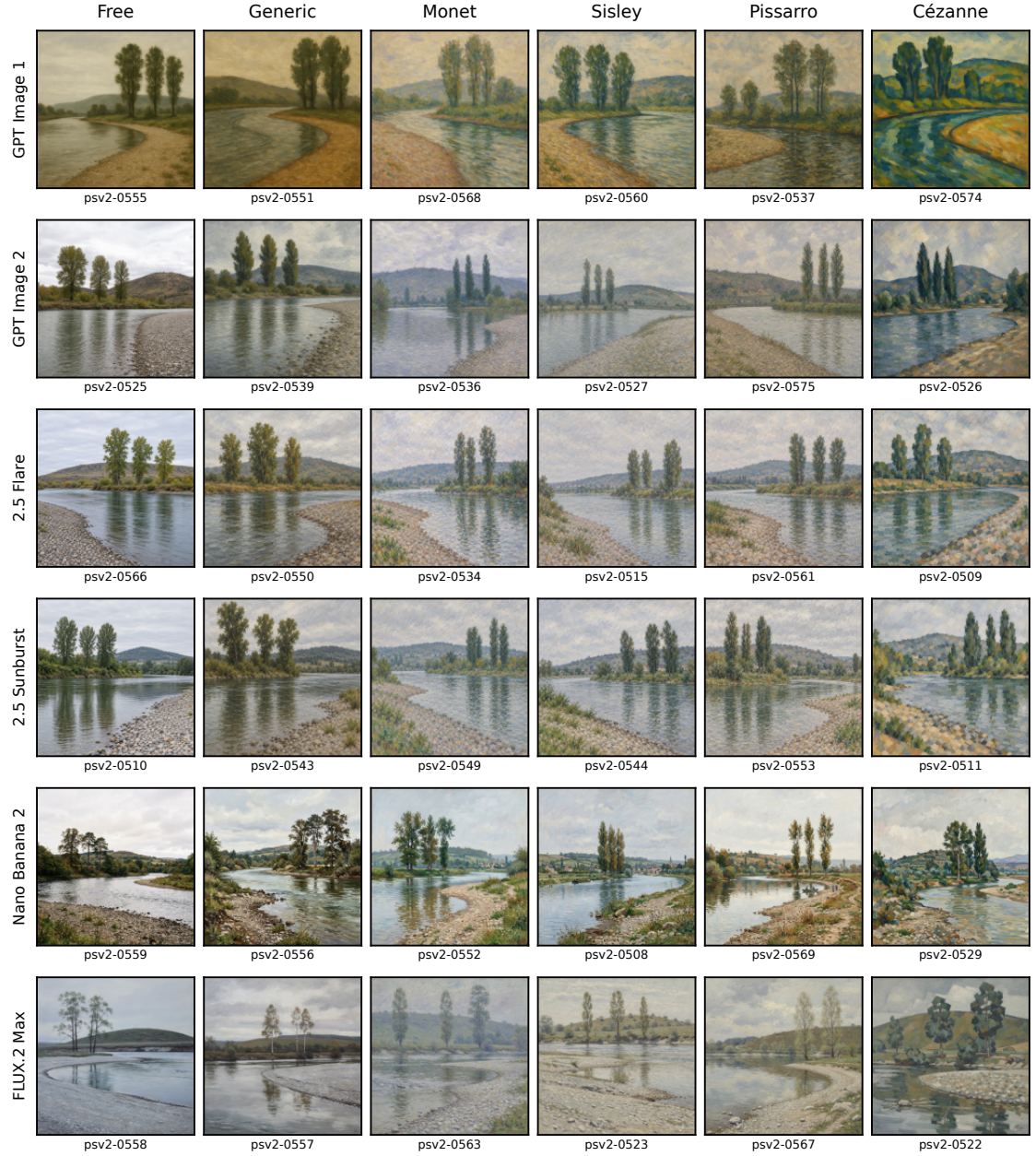


Figure 4: One common brief across all six models and clauses, first repeat. Every thumbnail carries its stable request identifier. Rows compare recorded configurations; columns compare clauses. Selection is deterministic and post-result, without appearance-based curation. No human evaluation is inferred from this panel.

E Complete model and painter summaries

Table 6 reports the full model-comparison family; Table 7 retains distributional outcomes for every painter and model. The latter do not inherit the primary model-contrast significance tests. Their energy values concern empirical distributions of 28 named or generic outputs against the recorded reference panel.

Model A	Model B	$D_A - D_B$ [simultaneous interval]
GPT Image 1	GPT Image 2	0.346 [−0.400, 1.093]
GPT Image 1	2.5 Flare	−0.165 [−1.027, 0.697]
GPT Image 1	2.5 Sunburst	−0.069 [−1.181, 1.043]
GPT Image 1	Nano Banana 2	0.463 [−0.602, 1.528]
GPT Image 1	FLUX.2 Max	0.771 [−0.037, 1.580]
GPT Image 2	2.5 Flare	−0.512 [−1.150, 0.126]
GPT Image 2	2.5 Sunburst	−0.415 [−1.181, 0.350]
GPT Image 2	Nano Banana 2	0.116 [−0.922, 1.155]
GPT Image 2	FLUX.2 Max	0.425 [−0.161, 1.011]
2.5 Flare	2.5 Sunburst	0.096 [−0.286, 0.478]
2.5 Flare	Nano Banana 2	0.628 [−0.283, 1.540]
2.5 Flare	FLUX.2 Max	0.937 [0.256, 1.618]
2.5 Sunburst	Nano Banana 2	0.532 [−0.436, 1.500]
2.5 Sunburst	FLUX.2 Max	0.840 [0.058, 1.623]
Nano Banana 2	FLUX.2 Max	0.309 [−0.847, 1.464]

Table 6: All 15 paired model comparisons in the prespecified inferential family. A negative contrast favors model A on the stated feature-geometry target. Flare and Sunburst denote the corresponding GPT Image 2.5 models.

Model	Monet	Sisley	Pissarro	Cézanne
GPT Image 1	−2.515 / 0.479	−2.739 / 0.694	−2.628 / 0.603	−1.336 / 0.598
GPT Image 2	−0.602 / 0.283	−1.092 / 0.312	−0.195 / 0.303	−0.373 / 0.303
2.5 Flare	0.962 / 0.269	−0.008 / 0.371	0.429 / 0.321	−1.013 / 0.275
2.5 Sunburst	0.267 / 0.248	−0.718 / 0.274	−0.091 / 0.273	−1.653 / 0.318
Nano Banana 2	−0.513 / 0.829	−1.381 / 0.708	−0.741 / 0.572	−1.102 / 0.877
FLUX.2 Max	−0.707 / 0.340	−0.886 / 0.416	−1.080 / 0.570	−1.431 / 0.622

Table 7: Complementary outcomes for all four painters: each cell gives named-minus-generic reference energy / generated-to-reference total variance. These are descriptive empirical distribution summaries, without additional tests. Lower energy change favors naming over the generic clause; a variance ratio below one indicates a narrower generated feature distribution.

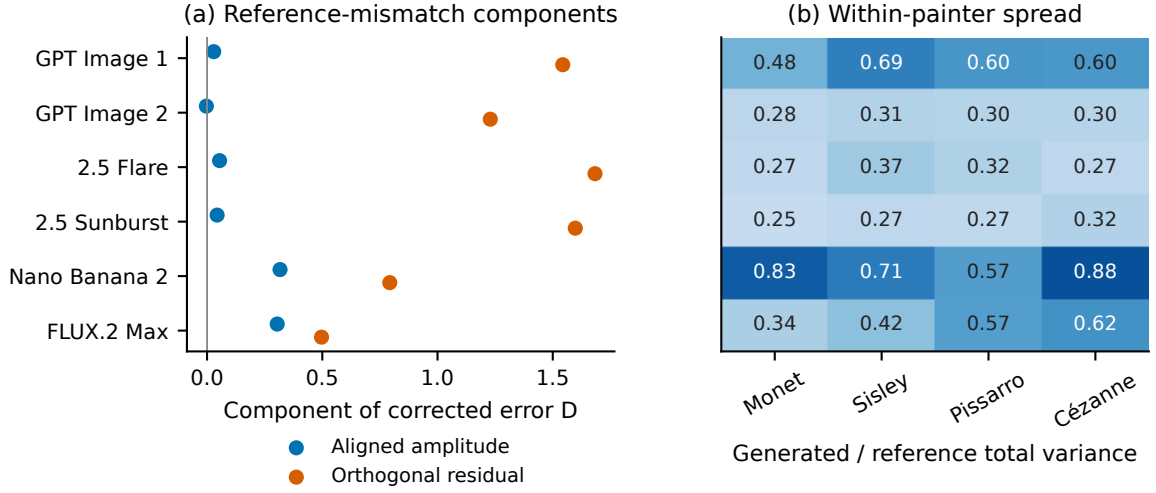


Figure 5: Complementary descriptive summaries. (a) Reference-aligned amplitude error and the residual orthogonal to the stacked reference configuration. (b) Generated/reference variance ratios for every model and painter. Total spread includes variation across fixed briefs and is not a mode-collapse measure.

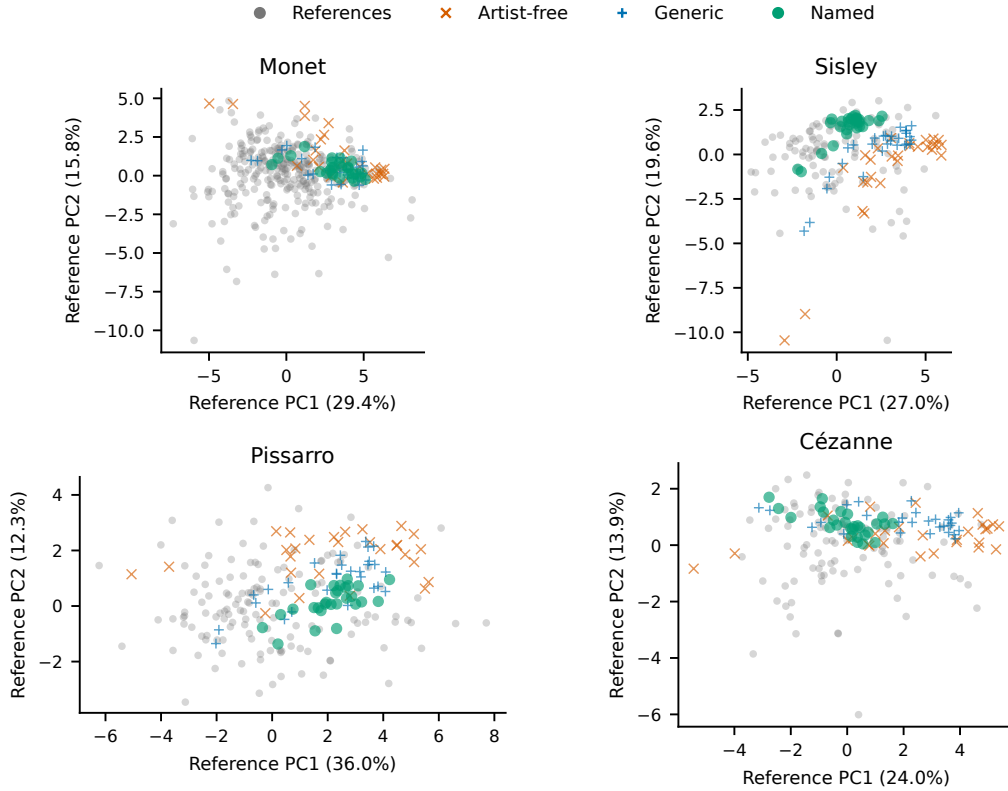


Figure 6: All four painters' reference and generated distributions, illustrated with GPT Image 2.5 Sunburst. This model illustration was selected before the primary feature outcomes were inspected. Each panel uses principal components fitted only to that painter's reference works, so axes differ between painters. The 14 generated briefs and two repeats have a different content distribution from the historical panel. Projections are descriptive; numerical summaries use all 31 coordinates.

F Supporting interventions

These earlier cohorts address related but different questions (Table 8). They are not pooled with the six-model experiment or treated as independent-investigator replications.

Cohort	Comparison	Role in this paper
Four-painter exploration	649 references; 1,920 generated outputs	Describes reference/generated separation and motivates the controlled test.
Controlled naming	1,006 outputs; 38 Monet and 32 Cézanne references	Six named/free and two detailed/short prompt contrasts; subsequent transformation and conditional-variance diagnostics.
Palette interventions	Two separate 192-output collections	Named-minus-generic chroma response; no pooling across collections.
Generic-clause comparison	96 Cézanne/generic outputs on 24 new scenes	Tests a proximity gain beyond a generic painting clause.
Fixed-map transfer	240 FLUX/Cézanne outputs on 12 new scenes	Prospective counterexample to the historical opposing energy/prediction ordering.

Table 8: Supporting evidence is kept separate from the new six-model experiment. Different prompts, panels and finite-sample targets do not form one pooled test.

F.1 Proximity and spread under naming

The four-painter exploration used 1,536 named and 384 artist-free outputs, three prompt methods and 16 scenes. Its 24 model-painter-method cells had trace ratios .206–.376 and grouped generated/reference discrimination accuracy .940–.983. Pooled named energy medians for Monet, Sisley, Pissarro and Cézanne were 2.068, 1.678, 1.975 and 2.142; corresponding artist-free medians were 1.732, 1.878, 1.854 and 1.657. Only Sisley had a lower pooled named median. Matched-size real/real medians were .193, .215, .164 and .199. These post-result summaries describe that exploration, not a general naming effect.

The later controlled panel selected Monet and Cézanne after exploration. It crossed 24 scenes with three models, two clauses and three repeats; 142 short-scene outputs supplied two prompt-length comparisons. Reference panels contained 38 Monet and 32 Cézanne works, with water/built/land masses based on maintainer-run LLM visual coding. Named-minus-free energy changes were $-.846, -1.818$ for Nano Banana 2, $-.625, -.896$ for FLUX, and $-.234, -.056$ for GPT Image 2 (Monet, Cézanne). The first four rejected in the original family of eight. A separate 72-output FLUX collection repeated both directions ($-.695, -.952$) with rejection in its own four-endpoint family.

A standalone 96-output Cézanne comparison on 24 new scenes reduced energy relative to the generic clause by 1.195 ($p = .00001$, fixed $\alpha = .025$), with named/generic trace ratio .552. It randomized clauses within 48 two-position blocks, used reference class masses (3, 11, 18)/32, and tested a sharp no-effect null with 99,999 within-block sign draws and plus-one correction. This tests clause effects, not equality of energy alone. A generic clause therefore did not reproduce the full proximity gain in that panel, but proximity does not establish correct four-artist contrasts. Likewise, lower total spread does not establish loss of scene information: controlled FLUX/Monet held-repetition scene retrieval rose from 44.4% to 59.7% under naming. This measures relative feature distinguishability, not verified prompt adherence.

F.2 Palette responsiveness and a contrary transfer result

Two separate 192-output palette collections used six scenes, four repeats, muted/vivid instructions and free, generic, Monet and Cézanne clauses. The primary name-minus-generic chroma-

response interactions were unresolved: $-.284, -.018$ in the first collection and $-.159, .037$ in the later one. Their respective adjusted intervals spanned zero; the original family had two interactions and the later family included two additional FLUX endpoints. Secondary generic-minus-free responses were $-.853$ and $-.891$, with nominal 95% intervals excluding zero. Thus generic painting can attenuate measured palette response without an identified additional name effect. Two palette levels do not distinguish reduced gain from saturation or an operating-range shift, and unresolved interactions do not demonstrate equivalence.

Historical generated-only moment maps were fitted on training scenes and assessed on held-out scenes. Actual naming had lower reference energy than translation plus isotropic scaling in all six model-painter averages. Adding scale improved named-mean prediction in all six while worsening reference energy in five. That ordering did not transfer to a prospective 240-output FLUX/Cézanne comparison with 12 new scenes and ten repeats, using the original fits unchanged. Translation/scaling improved both energy and repeat-corrected prediction error relative to translation: $-.390 [-.509, -.271]$ and $-5.047 [-7.880, -2.214]$, with approximate simultaneous jackknife intervals. The achieved prediction-error half-width exceeded its planning criterion. This contrary result is retained without extra sampling; it rules out treating the historical opposing ordering as a universal mechanism. These fitted named/free maps are distinct from the review-driven artist-contrast scalar counterfactual in Table 3.

G Collection provenance and computational scope

Historical GPT Image 1 and GPT Image 2 labels identify the requested model names. A later technical negative control showed that the earlier image interface also accepted an invented image-model identifier. This does not show that valid names select the same underlying model, but successful request acceptance alone cannot authenticate their separation. Historical contrasts are therefore interpreted as recorded request-group comparisons. The current experiment uses explicit documented model identifiers and fixed routes; it does not reuse historical images as verified examples of the new variants. All current outputs decode to the requested square dimensions, but their response bodies do not echo model or quality fields. Labels therefore denote the documented request configurations, without independent checkpoint attestation. Underlying checkpoints and training corpora are not available for inspection.

The present experiment excludes technical probes and an earlier attempt closed before feature measurement because model selection could not be qualified. No image is shared between these collections. The earlier incomplete generic-clause cohort is likewise not pooled with its complete 96-image successor. Exact assignments, attempt outcomes and measurement exclusions are retained in the versioned manifests; selection was not based on visual success.